



OctoSlave

Autonomous AI research & coding assistant — powered by e-INFRA CZ, NVIDIA NIM, or your own local GPU.

- 3 working modes
- science + lab
- 35+ built-in tools
- fully local option

What OctoSlave Can Do

OctoSlave is a single autonomous agent that plans, executes, verifies, and iterates. Point it at a task — code, research, analysis, or a full multi-agent investigation — and it works end-to-end until the job is done.

1

Code & engineering
Explores repos, writes/refactors/debugs code, runs tests, builds and deploys. Supports Python (uv), JS, Go, Rust, C/C++, and more.
agentic loop

2

Web research
DuckDuckGo search, full-page & PDF extraction, and an intelligent BFS site crawler. Synthesises findings across sources.
live web

3

Bio & chemistry
Direct REST access to UniProt, PubChem, ChEMBL, RCSB PDB, AlphaFold, GEO, ENA + RDKit for molecule analysis.
domain tools

4

Self-verifying
Writes a numbered plan before acting, grades its own completion (DONE / PARTIAL / FAILED), and nudges out of repeated failures.
self-check

5

Autonomous Lab
Self-organizing research team — a Director builds up to 10 specialists, a Critic gates the plan, produces an HTML report.
multi-agent

6

MCP & remote
Plugs in 21+ external tools (Git, GitHub, browser, DBs) via MCP. Runs tools on remote SSH hosts with GPU access.
extensible

How to Use It

Get started in 3 steps

1

Install
Download the bundled installer — no Python required. macOS, Windows, and Linux (AppImage) all available.

2

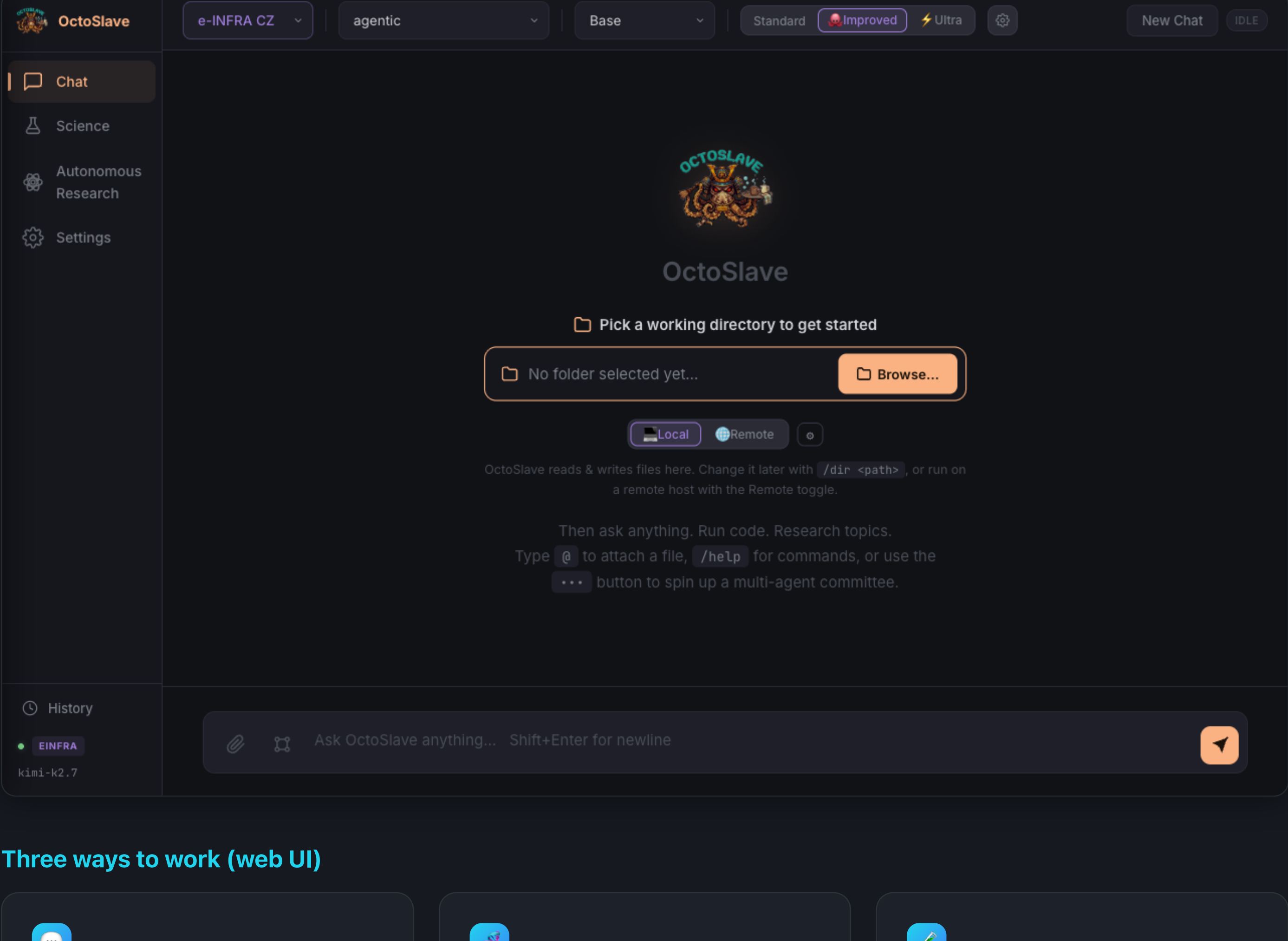
Configure
Add an e-INFRA CZ or NVIDIA NIM key — or go fully local with Ollama for complete privacy.

3

Run
Type a task and watch OctoSlave plan, execute, and report back — interactively or one-shot.

Web UI (browser)

Start the visual interface in one command — plan, execute, and review results in your browser. Works locally (127.0.0.1:7860) or share the link to let others join.



Three ways to work (web UI)

Chat
A single agent for everyday coding & analysis. Optional Improved / Ultra council for harder problems.

Science
Conversational research orchestrator. Spins up specialists on demand, presents plots & tables inline for refinement, curates data into FAIR datasets.

Autonomous Research
Self-organizing team as a batch pipeline — a Director assembles specialists, a Critic gates the plan, the team writes an HTML report.

Model Selection

3 BACKENDS

13+ KNOWN MODELS

1M MAX CONTEXT

0 API KEYS NEEDED (LOCAL)

Backends — pick your trade-off

Any custom OpenAI-compatible provider can be connected — just set the endpoint and API key in `ots config`. Built-in backends include e-INFRA CZ, NVIDIA NIM, and Ollama.

e-INFRA CZ default
Czech academy cloud. Best quality, free for Masaryk University researchers. Requires an API key.
`kimi-k.3`, `deepseek-v3.2-thinking`, `qwen3-coder-30b`

NVIDIA NIM
Cloud inference for frontier open-weight models via an OpenAI-compatible API.
`nvldia/nemotron-3-super-120b-a12b`

Ollama private
Run models locally on your GPU. Zero API keys, zero data leaves your machine.
`llama3.1:8b` and many more

Recommended models (e-INFRA CZ)

Goal	Model	Best for
Start here	<code>kimi-k.3</code>	Best all-round agent
Hard problems	<code>glm-5.2</code>	Chain-of-thought, complex reasoning
Code generation	<code>laguna-s2.1</code>	Strongest coder
Writing-heavy	<code>gpt-oss-120b</code>	Large context, clean prose
Fast reading	<code>qwen3.5-122b</code>	Large research datasets

`ots models` lists everything available on your backend. Override per-task with `--m MODEL` or switch live with `/model NAME` in a session.

Working Modes

StandardImprovedUltra

A single capable agent — fast and direct
One model drives the entire task. Perfect for everyday coding, quick questions, and tasks where a single strong model is enough. Low token cost and lowest latency.
`$ ots run "build a todo API"`

Science & Lab Options

For researchers who need real, reproducible results — not just summaries. Conversational orchestration and autonomous multi-agent teams, with live HPC offloading and FAIR data curation.

Science tab /science
Chat-first research orchestrator. You stay in the loop — every plot, table, and dataset appears inline for comment-driven refinement.

- Spins up specialists on demand (Structural Biologist, Data Wrangler, ...)
- Presents plots & tables inline — edit or regenerate specific outputs
- Full bio/chem toolbox: UniProt, PubChem, ChEMBL, PDB, AlphaFold, GEO, ENA, RDKit
- Offloads long jobs to Slurm/PBS or detached background processes
- Curates messy data into FAIR datasets (`datapackage.json`)
- Finds the most relevant current literature (Europe PMC)

web-only

Autonomous Research tab /Lab
Self-organizing research team as a batch pipeline. Hand off a whole problem and let a team run it end-to-end.

- Director builds a custom team of up to 10 specialists per task
- Critic challenges every plan before implementation
- Runtime tool foundry — agents build new tools mid-run
- Add teammates or connect MCP servers on demand
- Produces a self-contained HTML report
- Human-in-the-loop: step-mode approval gates or full autonomy

batch pipeline

Prompt profiles — match the agent to the task

Profile	Best for	Use when...
base default	General tasks, coding, mixed research	Full research workflow built in
coder coding	Pure code — refactor, implement, test	Skip research preamble, go straight to code
analyst data	Datasets, statistics, visualisations	CSV/JSON analysis with pandas, scipy, matplotlib
cryouncle cryo	Cryo-EM single-particle workflow	Connects directly to CryoSPARC

Extend & connect

MCP tools
Plug in ~21 curated servers: Git, GitHub, Playwright browser, SQLite/Postgres, Slack, Notion, Google Drive, and more.

```
$ /mcp registry # browse catalog
$mcp install github # one-click install
$mcp add mysrv "uvx ..." # custom server
```

Remote execution
Run tools on a remote SSH host — for remote CPUs, compute clusters, or data that lives on a server. Add once, switch instantly. Below is the Web UI's connection panel — click the highlighted button to switch the active backend to your remote PC.

```
$ ots run "train model!" --remote gpu01
/remote add | list | <id> | local
```

Ready to turn your ideas into real results?

Download OctoSlave — free for Masaryk University researchers.
Try it locally, in the cloud, or via the web.

Download the installer